



<- This means this lab is on your local computer

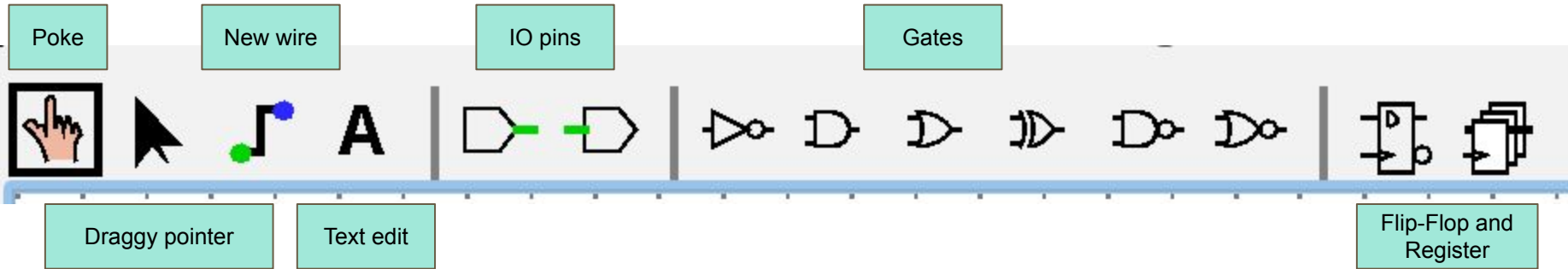
Lab 5: Logisim

CS61C SU26

Topics

- Logisim tools
- Basic logic gates in Logisim
- Advanced (and useful) Logisim components
- Wire colors in Logisim

Logisim Basics



- Don't forget about debugger finger!!

AND Gate

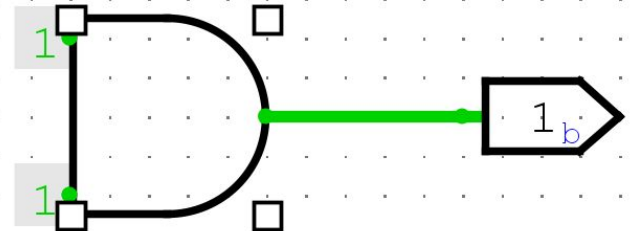
2 - input AND gate



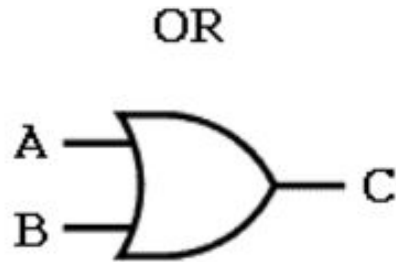
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

AND - logisim

Properties	State
AND Gate (120,360)	
FPGA supported:	Supported
Facing	→ East
Data Bits	1
Gate Size	Medium
Number Of Inputs	2
Output Value	0/1
Label	
Label Font	SansSerif Bold 16
Negate 1 (↑ Top)	No
Negate 2 (↓ Bottom)	No



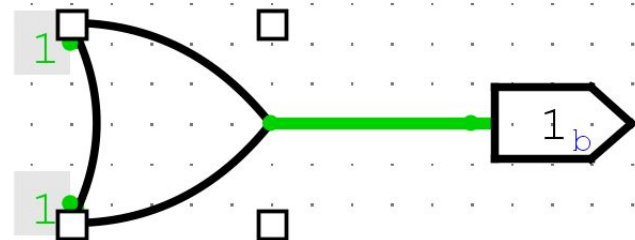
OR Gate



Inputs		Output
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

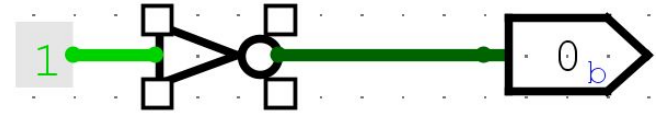
OR - logisim

Properties	State
OR Gate (120,360)	
FPGA supported:	Supported
Facing	→ East
Data Bits	1
Gate Size	Medium
Number Of Inputs	2
Output Value	0/1
Label	
Label Font	SansSerif Bold 16
Negate 1 (↑ Top)	No
Negate 2 (↓ Bottom)	No



NOT Gate

Properties	State
NOT Gate (110,320)	
FPGA supported:	Supported
Facing	→ East
Data Bits	1
Gate Size	Wide
Output Value	0/1
Label	
Label Font	SansSerif Bold 16



Adder

Design Simulate

+ VH DL ↑ ↓ ✕

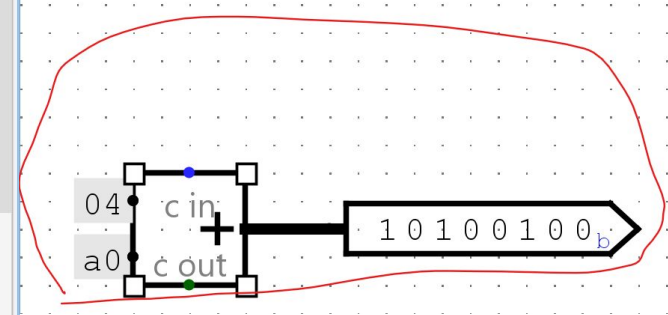
- ▶ Gates
- ▶ Plexers
- ▼ Arithmetic
 - +** Adder
 - Subtractor
 - × Multiplier
 - ÷ Divider
 - x Negator
 - Comparator
 - Shifter
 - # Bit Adder
 - ? Bit Finder
 - + Floating Point Adder
 - Floating Point Subtractor
 - × Floating Point Multiplier
 - ÷ Floating Point Divider
 - x Floating Point Negator

Properties State

Adder (80,340)

FPGA supported: Supported

Data Bits 8



Shifter

Design Simulate

+ VH ↑ ↓ ✗

▶ Plexers

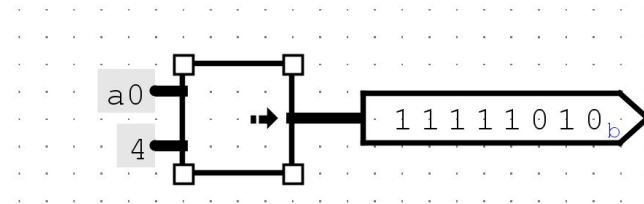
▼ Arithmetic

- + Adder
- Subtractor
- × Multiplier
- ÷ Divider
- x Negator
- Comparator
- Shifter
- # Bit Adder
- ? Bit Finder
- + Floating Point Adder
- Floating Point Subtractor
- × Floating Point Multiplier
- ÷ Floating Point Divider
- x Floating Point Negator
- Floating Point Comparator

Properties State

Shifter (120,330)

FPGA supported:	Supported
Data Bits	<u>8</u>
Shift Type	<u>Arithmetic Right</u>



Comparator

Design Simulate

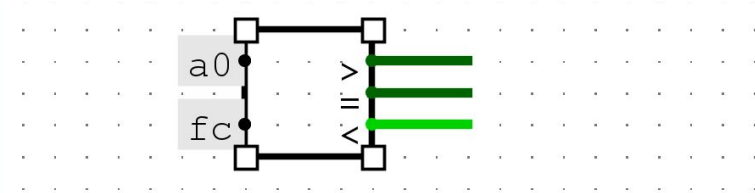
+ VH DL ↑ ↓ ✕ ✕

- ▶ Plexers
- ▼ Arithmetic
 - + Adder
 - Subtractor
 - × Multiplier
 - ÷ Divider
 - * Negator
 - Comparator**
 - Shifter
 - # Bit Adder
 - ? Bit Finder
 - + Floating Point Adder
 - Floating Point Subtractor
 - × Floating Point Multiplier
 - ÷ Floating Point Divider
 - * Floating Point Negator
 - Floating Point Comparator

Properties State

Comparator (120,340)

FPGA supported:	Supported
Data Bits	8
Numeric Type	2's Complement



Register

Design Simulate

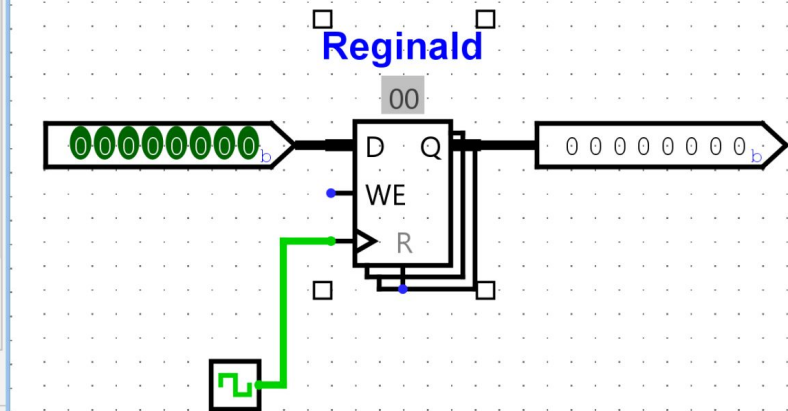
+ VH DL

- Wiring
- Gates
- Plexers
- Arithmetic
- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
- Input/Output
- TTL
- TCL

Properties State

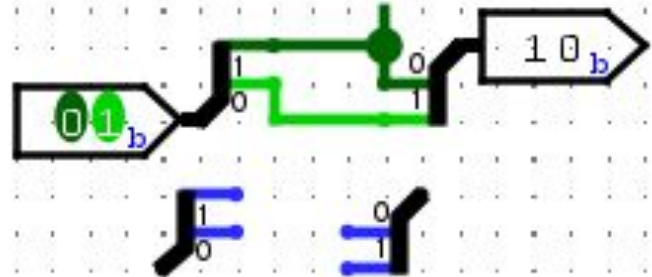
Register "Reginald"

FPGA supported:	Supported
Data Bits	8
Trigger	Rising Edge
Label	Reginald
Label Location	1 North
Label Font	SansSerif Bold 16
Show in Registers Tab	No
Appearance	Logisim-Evolution



Splitter (important!)

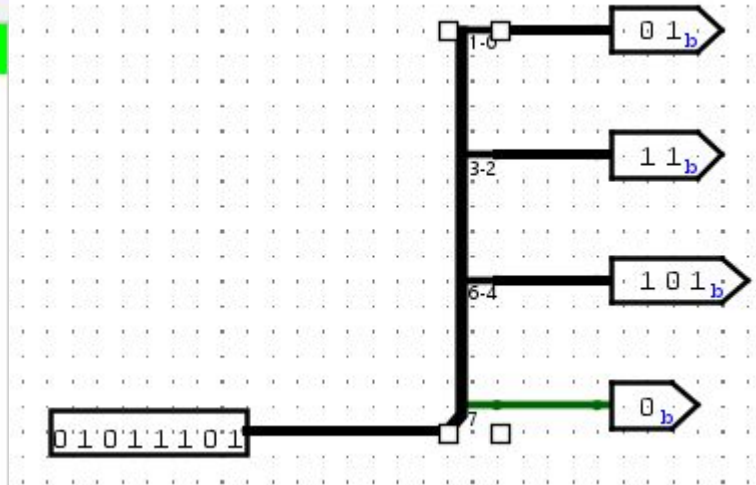
Splitter (270,270)	
FPGA supported:	Supported
Facing	← West
Fan Out	<u>2</u>
Bit Width In	<u>2</u>
Appearance	Left-handed
Spacing	1
Bit 0	<u>0 (↑ Top)</u>
Bit 1	<u>1 (↓ Bottom)</u>



Splitter cont.

Splitter (590,720)

FPGA supported:	Supported
Facing	→ East
Fan Out	4
Bit Width In	8
Appearance	Left-handed
Spacing	5
Bit 0	0 (↑ Top)
Bit 1	0 (↑ Top)
Bit 2	1
Bit 3	1
Bit 4	2
Bit 5	2
Bit 6	2
Bit 7	3 (↓ Bottom)



Tip: You can shift-click the bit rows to select multiple bits. Editing one bit's pins will then edit the pins for multiple bits at once! This can save lots of time.

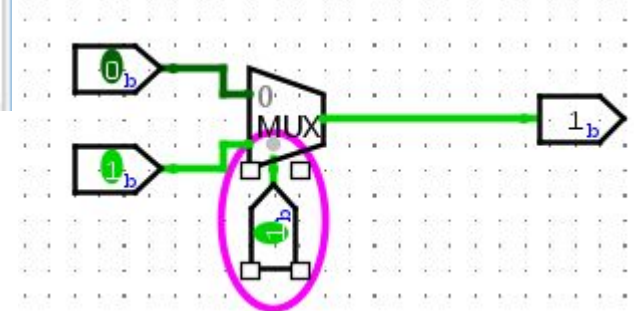
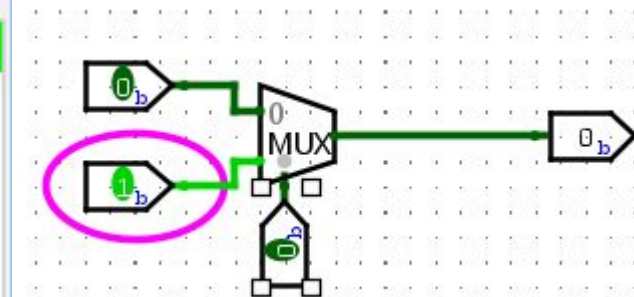
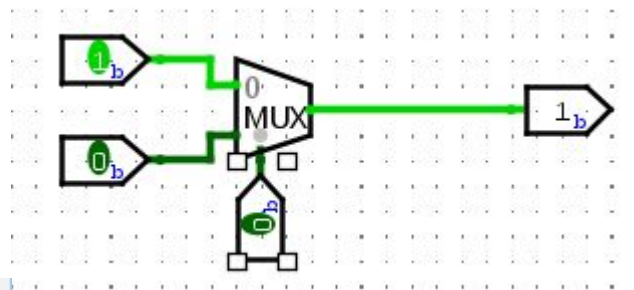
Mux

Multiplexer (180,430)

FPGA supported:

Supported

Facing	→ East
Gate Size	Wide
Select Location	Bottom/Left
Select Bits	1
Data Bits	1
Disabled Output	Zero
Include Enable?	No



Wire colors and their meanings (from spec)

Color	Meaning
Dark green	1-bit wire has a value of 0
Bright green	1-bit wire has a value of 1
Black	Multi-bit wire (many components have bit width attributes which can be configured in the attributes menu on the bottom left)
Red (values with EEEE)	The wire has multiple values on it (in this case, a 0 and 1 from the 2 inputs). Also, remember that a big circle appears at wire junctions.
Blue (values with UUUU)	The wire is floating (i.e. has no known value)
Orange	The wire is connected to components that have different bit widths. A simple example is a 1-bit input pin connected to a 2-bit output pin. In the shown example, there's a slightly hidden wire behind the MUX connecting the 2-bit lower data line to the 1-bit select line. Watch out for these!

Demo

- DEMO TIME